

Assignment 1

2026-09-15

```
tinytex::is_tinytex()
```

```
## [1] TRUE
```

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

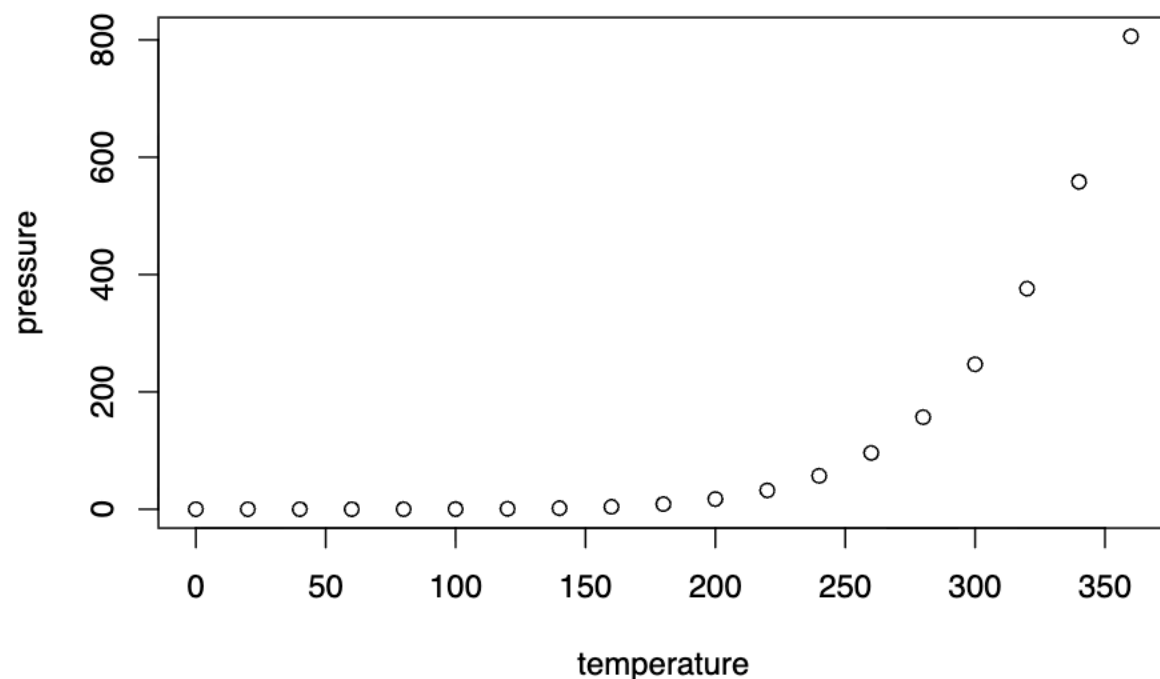
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0    Min.   : 2.00
##  1st Qu.:12.0    1st Qu.: 26.00
##  Median :15.0    Median : 36.00
##  Mean   :15.4    Mean   : 42.98
##  3rd Qu.:19.0    3rd Qu.: 56.00
##  Max.   :25.0    Max.   :120.00
```

Including Plots

You can also embed plots, for example:



```
summary(iris)
```

```
##   Sepal.Length   Sepal.Width   Petal.Length   Petal.Width
##   Min.    :4.300   Min.    :2.000   Min.    :1.000   Min.    :0.100
##   1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
##   Median :5.800   Median :3.000   Median :4.350   Median :1.300
##   Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
##   3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
##   Max.    :7.900   Max.    :4.400   Max.    :6.900   Max.    :2.500
##      Species
##   setosa    :50
##   versicolor:50
##   virginica :50
##
##
##
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

I created this cell to do a log transformation to decrease the skewness of the data

```
iris$Sepal.Length_log <- log(iris$Sepal.Length)

head(iris[, c ("Sepal.Length", "Sepal.Length_log")])
```

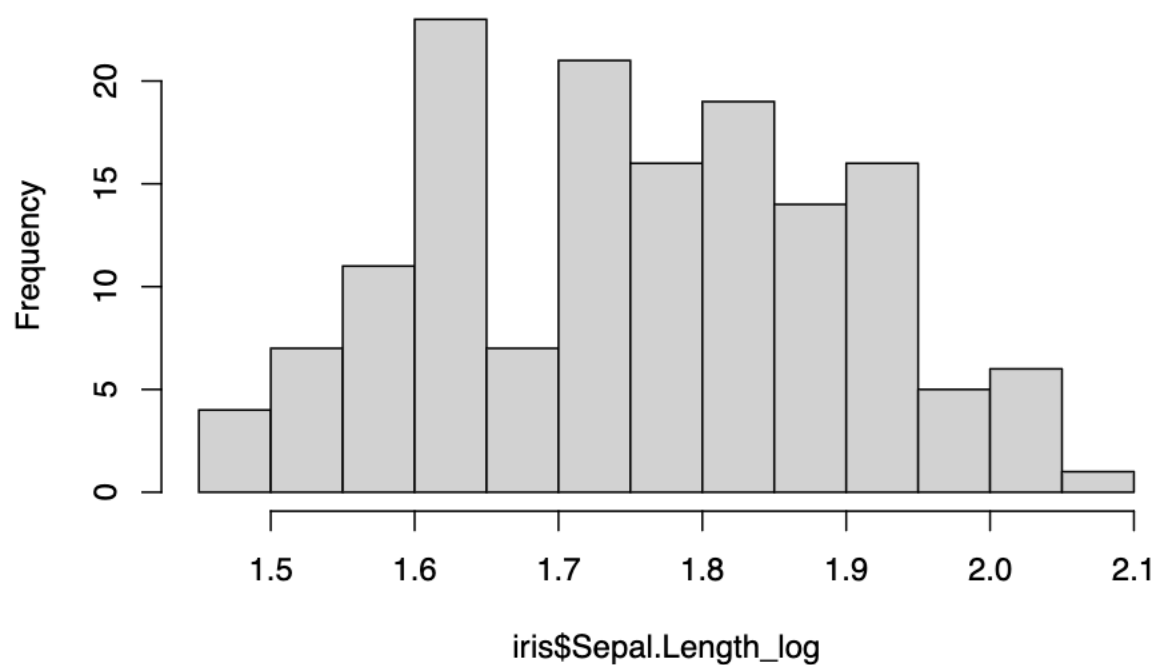
```
##   Sepal.Length Sepal.Length_log
## 1          5.1          1.629241
## 2          4.9          1.589235
## 3          4.7          1.547563
## 4          4.6          1.526056
## 5          5.0          1.609438
## 6          5.4          1.686399
```

```
summary(iris)
```

```
##   Sepal.Length   Sepal.Width   Petal.Length   Petal.Width
##  Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
## 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
##  Median :5.800   Median :3.000   Median :4.350   Median :1.300
##  Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
## 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
##  Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
##      Species   Sepal.Length_log
##  setosa      :50   Min.   :1.459
##  versicolor:50   1st Qu.:1.629
##  virginica  :50   Median :1.758
##                      Mean   :1.755
##                      3rd Qu.:1.856
##                      Max.   :2.067
```

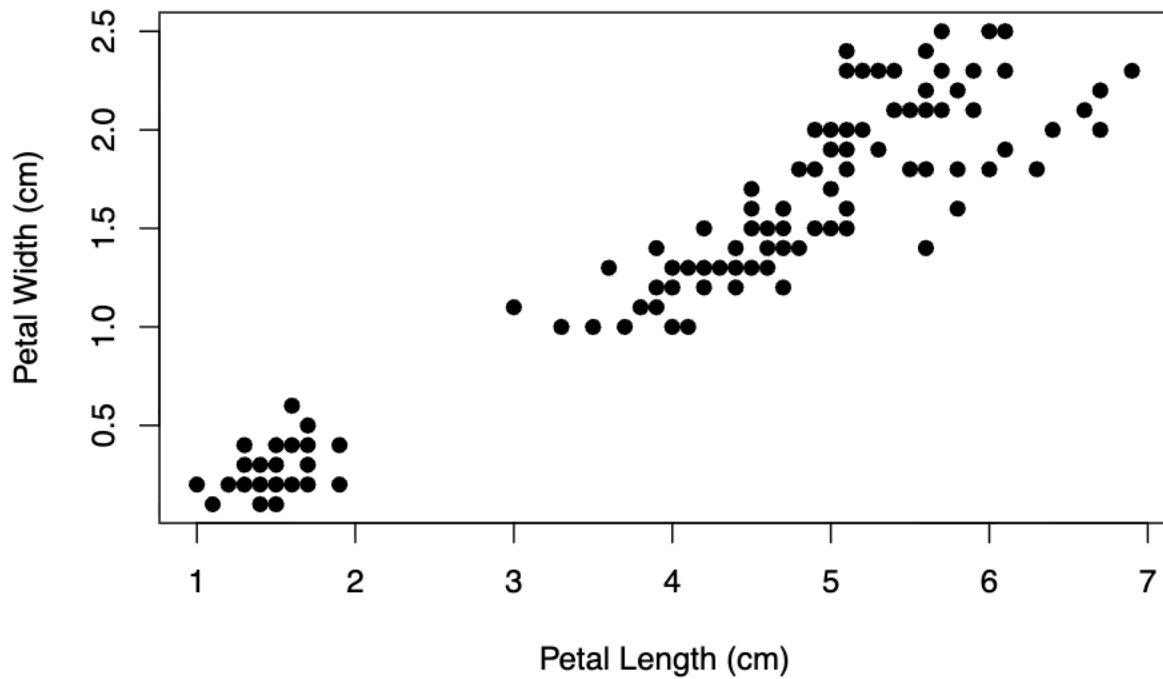
```
hist(iris$Sepal.Length_log)
```

Histogram of iris\$Sepal.Length_log



```
# Create a scatterplot of Petal Length vs. Petal Width
plot(iris$Petal.Length, iris$Petal.Width,
     main = "Petal Length vs. Petal Width",
     xlab = "Petal Length (cm)",
     ylab = "Petal Width (cm)",
     pch = 19)
```

Petal Length vs. Petal Width



```
# Create a scatterplot with different colors for each species
plot(iris$Petal.Length, iris$Petal.Width,
     main = "Petal Length vs. Petal Width by Species",
     xlab = "Petal Length (cm)",
     ylab = "Petal Width (cm)",
     col = as.factor(iris$Species),
     pch = 19)

legend("topleft",
      legend = levels(as.factor(iris$Species)),
      col = 1:3,
      pch = 19)
```

Petal Length vs. Petal Width by Species

